Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2021	ENGI3391-WE01

Title: **MEng/BEng:** Control and Signal Processing 3		
Time Allowed:	2 hours	
Additional Material provided:	Yes - Formula sheet, included as part of exam paper	
Materials Permitted:		
Calculators Permitted:	Yes	Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: Yes		

Instructions to Candidates:	Answer ALL questions. All relevant workings must be shown.

Revision:	

Section A

Q.1

Consider a periodic signal which is expressed in the form of Eqn. 1 as

$$x(t) = \sum_{k=-3}^{3} a_k e^{jk2\pi t} \quad \text{(Eqn. 1)}$$

where

$$a_0 = 1,$$
$$a_1 = a_{-1} = \frac{1}{4},$$
$$a_2 = a_{-2} = \frac{1}{2},$$
$$a_3 = a_{-3} = \frac{1}{3}$$

(a) Determine the fundamental frequency of the signal $x(t)$.

[10%]

(b) Using Euler's identities rewrite $x(t)$ in trigonometric form.

[20%]

(c) Sketch both the single sided and the double sided spectrum of the signal $x(t)$.

[20%]

(d) Suppose that the signal $x(t)$ is applied to a linear time invariant system with impulse response $h(t)$ in the form of Eqn. 2 as

$$h(t) = e^{-t}u(t) \quad \text{(Eqn. 2)}$$

where $u(t) = \begin{cases} 1 & for\ t \geq 0 \\ 0 & elsewhere \end{cases}$

(i) Derive an expression for the magnitude and phase of the frequency response of the system $h(t)$.

[20%]

(ii) Sketch the magnitude of the single sided spectrum output of the system due to the input $x(t)$ in Eqn. 1.

[20%]

(e) Give an example of a nonlinear system.

[10%]

Continued

Q.2

(a) A signal with frequency components between 1 MHz and 4 MHz is to be sampled using an analogue to digital converter. Determine the specifications in terms of clock rate and number of bits of a suitable analogue to digital converter for a minimum dynamic range of 48 dB.

[20%]

(b) Figure 2.1 gives the block diagram of a system.

(i) Write expressions for the outputs of $y(t)$, and $z(t)$ in terms of the input $x(t)$.

[15%]

(ii) Assuming $x(t)$ is a causal rectangular pulse of amplitude A and duration equal to *T*, sketch the output of the system, $z(t)$.

[10%]

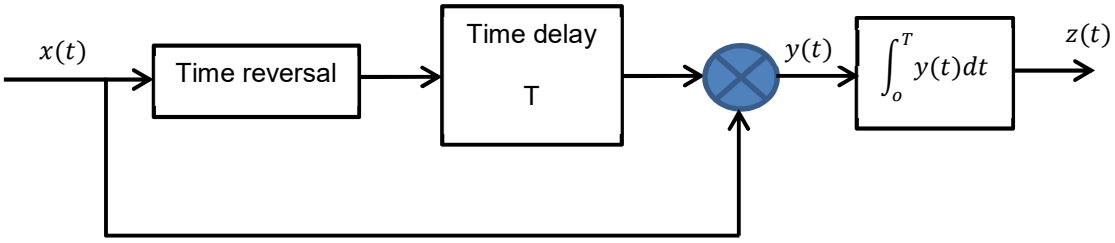


Figure 2.1 Block diagram of a system

(c) A linear time invariant system has an impulse response $h(t)$ given by Eqn. 3

$$h(t) = \delta(t) + 3\delta(t-T) + 5\delta(t-5T) \qquad \text{(Eqn. 3)}$$

Sketch its impulse response.

[10%]

(d) An input $z(t)$ given by Eqn. 4 is applied to the system described in Eqn. 3

$$z(t) = \begin{cases} t, & 0 < t < T \\ 0, & elsewhere \end{cases} \qquad \text{(Eqn. 4)}$$

(i) Find an expression for the output of the system $y(t)$ in response to $z(t)$

[20%]

(ii) Sketch the output of the system $y(t)$.

[15%]

(e) Give an expression for the overall impulse response of two linear time invariant systems assembled (i) in cascade, (ii) in parallel.

[10%]

Continued

Section B

Q.3

(a) State the purpose of compensation design in a control system, and the effects of adding a gain compensator, a phase-lag compensator, and a phase-lead compensator to the system.

[15%]

(b) A control system is shown in Figure Q3 with the process gain $(s) = \frac{10}{s^2+5s+6}$.

(i) Give the closed-loop transfer function of this system.
(ii) Calculate the closed-loop poles of this system.
(iii) Calculate the overshoot percentage of the system output when the system input is a step function.
(iv) Give the steady-state error of this system's output by using the final value theorem if the system input is a step function.

[20%]

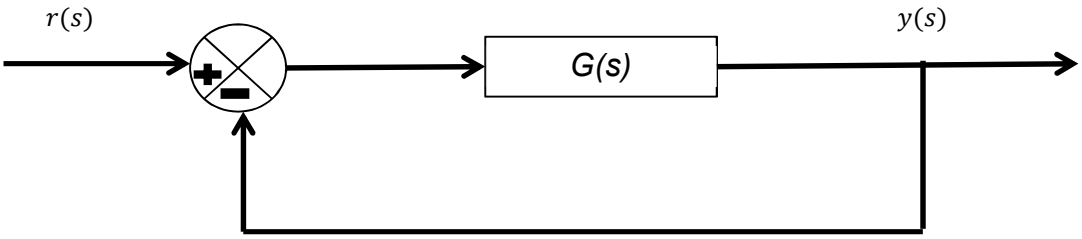


Figure Q3: A closed-loop control system with negative unit feedback.

(c) Sketch the root locus of this closed-loop system, and give the procedures and steps of the root locus plotting.

[15%]

(d) Design a practical phase-lead compensator *(hint: consider the format of* $K\frac{s+z}{s+p}$*)*, to meet two requirements at the same time: 1). The settling time of 2% error to its final value to be less than 1 second, and 2). The overshoot percentage to its final value to be less than 5%.

[30%]

(e) State the difference between a practical phase-lead compensator and an ideal phase-lead compensator in terms of design procedure, implementation and their effects on system output.

[20%]

Continued

Q.4

(a) State the meanings of gain margin and phase margin, and how you read them from Nyquist diagram and Bode diagram, respectively.

[20%]

(b) Consider a control system shown in Figure Q4 with the process gain $G(s) = \frac{20}{s^3+6s^2+11s+}$

(i) Ignoring the compensator, write the open-loop frequency response formula $GH(i\omega)$ in the complex number form of $a + ib$.

[10%]

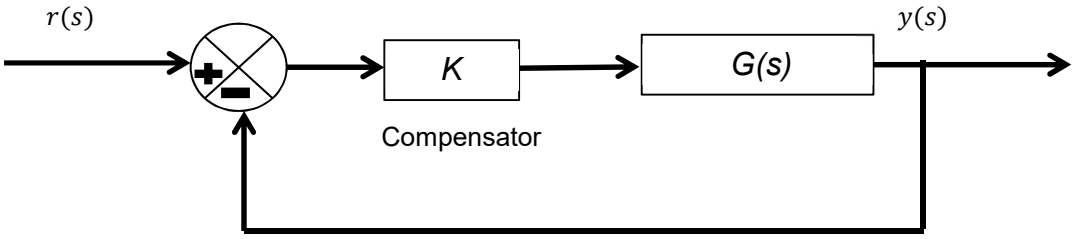


Figure Q4: A closed-loop control system with negative unit feedback.

(ii) Calculate its frequency response as ω varies, and present your data using a Nyquist diagram.

[20%]

(iii) Find the gain margin, the phase margin, and the phase cross-over frequency and the gain cross-over frequency.

[20%]

(iv) If the compensator is considered, what proportional gain *K* is required to achieve the phase margin of 80 degree? Analyse the effects of changing the phase margin from its original one in (iii) to 80 degree.

[30%]

L3 Control Formula Sheet

Standard second-order transfer function

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Closed-loop poles of under-damped second-order system

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

The overshoot percentage (%) of the standard second-order system

$$M_o = 100\exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

The settling time for a second-order system can be defined as

$$T_s = \frac{\text{Number of time constants}}{\zeta\omega_n}$$

The relation of number of time constants and settling error

Number of time constants	$e^{-\frac{t}{\tau}}$	Settling Error (%)
1	0.368	36.8
2	0.135	13.5
3	0.05	5
4	0.018	1.8
5	0.007	0.7

Defining $u(t)$ as the controller output and $e(t)$ as the error, the standard form of the PID algorithm is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{d}{dt}e(t)$$

Laplace form of the PID controller

$$G(s) = K_p + \frac{K_i}{s} + K_d s$$

Continued

7 Steps for Root Locus Sketching:

1). Write the characteristic function so that *K* is a multiplier *1+KG(s)=0.*

2). Factor *G(s)* in terms of *n* poles and *m* zeros:

$$G(s) = \frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{j=1}^{n}(s - p_j)}$$

3). Mark poles and zeros on the *s*-plane.

4). Determine the number of separate loci (lines)

5). Locate the segments of the real axis that are root loci

6). The remaining loci proceed to the zeros at infinity along asymptotes centered at σ and with angle *Φ*

$$\sigma = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n - m}$$

$$\phi = \frac{(2q+1)}{n-m}\pi,$$

where $q = 0, 1, \ldots, (n - m - 1)$

7). Plot remaining root locus.

Magnitude Criteria:

$$K = \frac{\prod_{i=1}^{n}|s - p_i|}{\prod_{j=1}^{m}|s - z_j|}$$

Angle Criteria:

$$\sum_{i=1}^{n}\phi_{p_i} - \sum_{j=1}^{n}\phi_{z_i} = (2q+1)\pi$$

Continued

Signal Processing Formula Sheet

Energy of a signal

$$E = \int_{-\infty}^{\infty} |f(t)|^2 dt$$

Power of a signal

$$P_{av} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt$$

Fourier series

$$x(t) = a_o + \sum_{k=1}^{\infty} \left[a_k \cos k\omega_o t + b_k \sin k\omega_o t \right]$$

where

$$a_o = \frac{1}{T} \int_T x(t)dt = dc\ term$$

$$a_k = \frac{2}{T} \int_T x(t) \cos k\omega_o t dt \qquad b_k = \frac{2}{T} \int_T x(t) \sin k\omega_o t dt$$

Alternative Fourier series

Trigonometric form:

$$x(t) = a_o + \sum_{k=1}^{\infty} \sqrt{a_k^2 + b_k^2} \cos(k\omega_o t + \phi_k)$$

where

$$\phi_k = -\tan^{-1} \frac{b_k}{a_k}$$

Complex exponential form

$$x(t) = \sum_{k=-\infty}^{\infty} d_k\, e^{jk\omega_o t}$$

where $d_k = \frac{1}{T} \int_T x(t).e^{-jk\omega_o t} dt$

and $d_k = \frac{1}{2}(a_k - jb_k)\ for\ k > 0\ and\ d_k = \frac{1}{2}(a_k + jb_k)\ for\ k < 0$

Continued

Fourier transform

$$\mathcal{F}\{x(t)\} = X(\omega)$$
$$= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

Alternatively it can be written in terms of the frequency in cycles per seconds or Hz as

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt$$

Inverse Fourier transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$$

or

$$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df$$

Fourier transform properties

1. Linearity

$$x_1(t) \leftrightarrow X_1(\omega), \text{and } x_2(t) \leftrightarrow X_2(\omega) \text{ then}$$
$$ax_1(t) + bx_2(t) \leftrightarrow aX_1(\omega) + bX_2(\omega)$$

2. Differentiation

$$y(t) = \frac{d^n x(t)}{dt^n} \leftrightarrow (j\omega)^n X(\omega)$$

3. Integration

$$y(t) = \int_{-\infty}^{t} x(\tau)d\tau \leftrightarrow \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)$$

4. Time scaling

$$x(bt) \leftrightarrow \frac{1}{|b|}X\left(\frac{\omega}{b}\right)$$

Continued

5. Time reversal

$$x(-\mathrm{t}) \leftrightarrow X(-\omega) = X^{*}(\omega)$$

6. Time delay

$$y(t) = x(\mathrm{t}-\mathrm{T}) \leftrightarrow X(\omega)\, \mathrm{e}^{-\mathrm{j}\omega \mathrm{T}}$$

7. Multiplication in time

$$x_1(\mathrm{t}).x_2(\mathrm{t}) \leftrightarrow \frac{X_1(\omega) * X_2(\omega)}{2\pi}$$

8. Multiplication in frequency (convolution in time)

$$X_1(\omega)X_2(\omega) \leftrightarrow x_1(\mathrm{t}) * x_2(\mathrm{t})$$

9. Modulation property

$$x(\mathrm{t})\cos(\omega_o t) \leftrightarrow \frac{1}{2}X(\omega - \omega_{\mathrm{o}}) + \frac{1}{2}X(\omega + \omega_o)$$

10. Duality property

If $x(t) \leftrightarrow X(\omega)$

Then $X(t) \leftrightarrow 2\pi x(-\omega)$

Discrete Fourier Transform

Discrete Fourier Series

DFS: $d_k = X[k] = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{\frac{-j2\pi kn}{N}}$

Inverse Discrete Fourier Series

IDFS: $x[n] = \sum_{k=0}^{N-1} d_k\, e^{\frac{j2\pi kn}{N}}$

Euler's identities

$$A\sin(\omega_o t) = \frac{A}{j2}\left(e^{j\omega_o t} - e^{-j\omega_o t}\right)$$

$$A\cos(\omega_o t) = \frac{A}{2}\left(e^{j\omega_o t} + e^{-j\omega_o t}\right)$$

Continued

$$A\cos(\omega_o t + \phi_o) = \frac{A}{2}\left(e^{j\omega_o t}e^{j\phi_o} + e^{-j\omega_o t}e^{-j\phi_o}\right)$$

$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$

Impulse function in continuous time domain

$\delta(t) = 0\ for\ t \neq 0\ and\ infinite\ at\ t = 0$

$$\int_{t=-\infty}^{\infty} \delta(t) = 1$$

Impulse function in discrete time domain

$\delta[n] = 1\ n = 0\ and\ 0\ elsewhere$

Unit step function in continuous time domain

$u(t) = 1\ for\ t \geq 0\ and\ 0\ t < 0$

Unit step function in discrete time domain

$u[n] = 1\ for\ n \geq 0\ and\ 0\ for\ n < 0$

Convolution integral

$$y(t) = x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)d\tau = \int_{-\infty}^{\infty} x_1(t-\tau)x_2(\tau)d\tau$$

Convolution sum

$$y[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k] = \sum_{k=-\infty}^{\infty} x_1[n-k]x_2[k]$$

Convolution with an impulse

$x(t) * \delta(t - t_o) = x(t - t_o)$

Correlation functions

Continuous time auto-correlation

Power signal: $\phi_{xx}(\tau) = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} x(t)x(t+\tau)dt$

Energy signal: $\phi_{xx}(\tau) = \int_{-\infty}^{\infty} x(t)x(t+\tau)dt$

Fourier transform of an impulse

Continued

$\delta(t) \leftrightarrow 1$

Inverse Fourier transform of an impulse

$\delta(\omega) \leftrightarrow 1/2\pi$

Roots of a second order polynomial

$ax^2 + bx + c = 0$

are given by

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Trigonometric identities

cos θ = cos(-θ) = sin(π/2 - θ)

sin θ = - sin(-θ) = cos(π/2 - θ)

tan θ = - tan(-θ) = cot(π/2 - θ)

sin 2θ = 2 cos θ sin θ

cos 2θ = $\cos^2\theta$ - $\sin^2\theta$ = 2 $\cos^2$ θ- 1 = 1 - 2 $\sin^2\theta$

tan 2θ = 2 tan θ/ (1 - tan2θ)

$$\sin\alpha.\sin\beta = \frac{1}{2}\{(\cos(\alpha - \beta) - cos(\alpha + \beta)\}$$

$$\cos\alpha.\cos\beta = \frac{1}{2}\{(\cos(\alpha - \beta) + cos(\alpha + \beta)\}$$

$$\sin\alpha.\cos\beta = \frac{1}{2}\{(\sin(\alpha - \beta) + sin(\alpha + \beta)\}$$

sin θ/2 = ± √[(1 - cos θ)/2]

cos θ/2 = ± √[(1 + cos θ)/2]

tan θ/2 = sin θ/(1 + cos θ)

$\cos^2$ θ+ $\sin^2$ θ = 1

$\sec^2$ θ - $\tan^2$ θ = 1

cosec^2 θ - $\cot^2$ θ = 1

$\sin^2$ θ = ½(1 - cos 2θ)

$\cos^2$ θ = ½(1 + cos 2θ)

Continued

$\tan^2 \theta = (1 - \cos 2\theta) /(1 + \cos 2\theta)$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\sin(A - B) = \sin A \cos B - \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\cos(A - B) = \cos A \cos B + \sin A \sin B$

$\tan(A + B) = (\tan A + \tan B)/(1 - \tan A \tan B)$

$\tan(A - B) = (\tan A - \tan B)/(1 + \tan A \tan B)$

$\sin A + \sin B = 2 \sin ½(A + B) \cos ½(A - B)$

$\sin A - \sin B = 2 \cos ½(A + B) \sin ½(A - B)$

$\cos A + \cos B = 2 \cos ½(A + B) \cos ½(A - B)$

$\cos A - \cos B = - 2 \sin ½(A + B) \sin ½(A - B)$

$\tan A + \tan B = \sin (A + B)/(\cos A \cos B)$

$\tan A - \tan B = \sin (A - B)/(\cos A \cos B)$

$\sin2A + \sin2B = 1 - \cos(A + B)\cos(A - B)$

$\sin2A - \sin2B = \sin(A + B)\sin(A - B)$

$\cos2A + \sin2B = 1 - \sin(A + B)\sin(A - B)$

$\cos2A - \sin2B = \cos(A + B)\cos(A - B)$

$\cos2A + \cos2B = 1 + \cos(A + B)\cos(A - B)$

$\cos2A - \cos2B = - \sin(A + B)\sin(A - B)$

For a triangle with sides a, b, c, and angles A, B, C opposite sides a, b and c respectively, the following relations hold.

$a2 = b2 + c2 - 2 b c \cos A$

$a/\sin A = b/\sin B = c/\sin C.$

$(a - b)/(a + b) = \tan ½(A - B)/ \tan ½(A + B)$

END